Spatially resolved spectral powder analysis: experiments and modeling

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Abstract

Understanding the behavior of light in granular media is necessary for determining the sample size, shape and weight when probing using fiber optic setups. This is required for a correct estimate of the active pharmaceutical ingredient content in a pharmaceutical blend via near-infrared spectroscopy.

Several strategies to describe the behavior of light in granular and turbid media exist. A common approach is the Monte-Carlo simulation of individual photons and their description using mean free path lengths for scattering and absorption. In this work, we chose a complementary method by approximating these parameters via real physical counterparts, i.e., the particle size, shape and density and the resulting chord lengths. Additionally, the wavelength dependence of refractive indices is incorporated.

The obtained results were compared with those obtained in an experimental setup that included the SAM-Spec® Felin probe head by Indatech for detecting spatially-resolved spectra of samples. Our method facilitates the interpretation of the acquired experimental results by contrasting the optical response, the physical particle attributes, and the simulation results.

Keywords

Near infrared spectroscopy, Powder technology, Pharmaceutical, Process Analytical Technology, Monte-Carlo

Introduction

In recent years near-infrared spectroscopy (NIRS) has gained much attention due to the ability to monitor processes online, as supported by the Process Analytical Technology (PAT) initiative.[1,2] The major advantage of NIRS is that it does not require sample preparation, making it possible to investigate solid and granular samples rapidly, without impacting the manufacturing processes. This explains why critical process parameters and critical quality attributes of pharmaceutical and food products are commonly determined via NIRS.[3–8]

Absorption and scattering

In a first approximation, recorded spectra are the result of two main factors: absorption and scattering.[9] The former can be used to identify and quantify chemical species based on their characteristic absorption behavior using the Beer-Lambert or the Kubelka-Munk theory.[10–12] The latter, which is affected by the physical properties of a sample, is sometimes used to establish correlations between the spectra and the particle size, bulk density and other structural properties.[13,14] The absorptive and scattering parts are inter-independent. A change in physical properties alters the observed absorption spectrum.[15,16] A change in a chemical composition modifies the physical properties of the sample (e.g., mixing various powder components with various particle sizes) and, ultimately, the scattering behavior.[9]

Efforts have been made to separate the absorptive and scattering parts of spectra. If one is only interested in the absorption, different spectral pretreatments can be used. However, both spectral parts can be separated (i.e., independently determined) by more elaborate means, e.g., by measuring different sample thicknesses or transmission and reflection spectra at the same time.[17,18]

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Another relevant topic in this context is the effective sample size, i.e. the volume of interaction leading to an acquired spectrum. Since scattering is the dominant contribution to light extinction in the NIR frequency region, the penetration depth of diffuse reflection based measurements can often be quite small (i.e., only a few particle layers), as photons are expelled from the specimen again, before having the chance to be absorbed. Resultantly, light intensity drops exponentially inside the specimen; the surface layer has a much greater influence on the measurement than the material inside the bulk and often dominates the acquired spectrum.[19] This effectively leads to a rather small investigated volume, as compared e.g. to wet-chemistry analytical methods.

Describing the behavior of light in granular media

Several approaches exist to understanding and describing the propagation of light in a granular material. To correctly describe the system, various length scales have to be used: the wavelength (10^{-6} m), the particle scale (10^{-4} m), the sample scale (10^{-3} m) and the bulk scale (10^{0} m).

An important parameter is the relative wavelength size λ and the particle size D, typically denoted as a non-dimensional factor $x = \pi D/\lambda$. For short wavelengths and large particles, interference phenomena can be ignored and the laws of geometrical optics hold. However, when the wavelength and the particle size are in the same order of magnitude, i.e., $x \approx 1$, various components of Mie theory have to be applied to describe the emerging light patterns. If the particles are small compared to the wavelength, the Rayleigh approximation can provide an analytical description of small-particle scattering.[20,21]

For describing a larger number of particles, various methods can be applied, depending on the field of interest:

- Exact analytical solutions via the Maxwell equations (i.e., Mie equations)[22–24]
- Stochastic methods (predominantly Monte-Carlo methods)[25–28]
- Geometrical optics (e.g. ray tracing)[29–31]
- Balance equations (known as radiative transfer equation)[16,32–34]

As near-infrared spectroscopy has gained a lot of attention in the pharmaceutical industry, the need for a detailed understanding of the behavior of radiation has increased, especially for products and intermediates like powder blends, granules, ribbons, and tablets. This is interconnected to the concept of scale of scrutiny. That is, it is vital to understand which volume in a process stream, sampled with NIR probes, determines the resulting signal. Even the concept of a sample volume may be misleading since the analysis with reflected penetrating radiation, especially in the case of near-infrared radiation, is weighed according to the depth of penetration. Hence, a defined volume is not sampled equally, but surface-near regions dominate the response.[35,36]

In the recent years, efforts have been made to simulate the propagation of light in pharmaceutical materials (i.e., in powders and tablets) and the effects of fiber optics via Monte-Carlo (MC) approaches. A study of single fiber optics and their effect on the penetration depth in biological tissues was performed by Kanick et al.[37] Various fiber optic probe designs were modeled by Papaioanno et al.[38] Arhallias et al. focused on the properties of the backscattering spot.[39] The penetration depth into pharmaceutical materials was investigated by Shi et al.[40] and Abrahamsson et al.[41] Pilorget et al. took into account the roughness and albedo of grains.[42] Okamoto et al. studied single powder layers with varying particle size,[43] and Tran et al. assessed the response of a fiber optical setup for various particle sizes.[44] An analytical solution for two-dimensional granular materials was provided by Sadjadi et al.[45] Svensson et al. dealt with porous media, consisting of scattering shapes and voids.[46,47] Research on the effect of spatial correlation in the media and the statistical description thereof can be found, e.g. in[48–51].

When light transmits a specimen, an experimental approach to determine the light paths inside a sample is to directly measure the flight time of photons. This was done, for example by Alerstam et al.[52]

Aim of this work

According to common approaches, like Monte-Carlo modeling of light transport in multilayered tissues (MCLML), the mean free path length between the absorbing and scattering events is chosen via elaborate sampling and calibration measurements, e.g., tissue phantoms.[53,54]

Here, in contrast, we treated the sample as a non-continuous medium consisting of particles and voids. Hence, the resulting chord lengths inside, and between particles, must deliver the same information as the particle size, shape and arrangement.[55] This avoids modelling a computationally costly and complex particle arrangement. Furthermore, the dependence of the chord length inside the particles, due to refraction at particle entrances, leads to a wavelength dependent behavior.

This method was applied with parameters obtained for several pharmaceutical powders. The calculation results were

Table 1. Particle attributes of powders used in the experiments. Multi-spherical in this context means that larger particles are granules of smaller spherical particles. Uncertainties were estimated by triplicate sampling.

Name	D_{10} (µm)	D_{50} (µm)	D_{90} (µm)	Bulk Density (kg/m^3)	Shape
Granulac 70	78	144	244	704 ± 14	spherical
Granulac 140	56	97	166	571 ± 21	spherical
Inhalac 70	147	225	299	684 ± 06	cubic
Inhalac 120	116	156	187	747 ± 10	cubic
Tablettose 70	90	196	362	596 ± 06	multi-spheric
Tablettose 100	66	158	360	587 ± 11	multi-spheric
SpheroLac 100	85	145	221	720 ± 04	spherical

compared to experimental measurements using a setup with a single illumination spot and detection spots positioned at an increasing distance. Using the SAM-Spec® Felin probe head (Indatech, Clapiers, France) a comparison in terms of spatially resolved intensity measurements was possible.

Thus, the presented approach has the advantage of using commonly measured and known particle parameters (i.e., the particle size distribution and bulk density) to simulate the sample's optical properties, without any need for parameter fitting.

Experimental Setup

Materials and material characterization

Materials with various particle sizes and shapes were investigated. The excipients were provided by Meggle GmbH (Wasserburg, Germany). All particles are chemically the same (α-lactose monohydrate) and only differ in their geometries. The relevant properties, as determined are listed in Table 1. These properties were used to model the log-normal particle size distribution and porosity (via $\varphi = 1 - \rho_{\text{bulk}}/\rho_{\text{true}}$, assuming that $\rho_{\text{true}} = 1546$ kg/m^3).[56]

For obtaining particle size distributions of the samples, measurements with an image based system were performed (QicPic, Sympatec GmbH, Clausthal-Zellerfeld, Germany). This system is based on a dispersion unit (Rodos) and a subsequent high speed imaging unit (QicPic), thus providing particle size and shape information.

Powder bulk density was determined during tests performed with a powder rheometer equipment (FT4 Powder Rheometer, Freeman Technology, Tewkesbury, United Kingdom).

The reason for the different particle properties and shapes is found in their preparation steps. Whereas Inhalac is only sieved, thus pertaining the more edgy shape, Granulac is milled, thus abrading sharp edges, leading to the more round appearance. SEM images were taken of the powder samples showing their difference, based on about 100 particles. Further hints about their shape can be gained by taking a look at the measured sphericity distributions, and images obtained during the particle size measurement. Nonetheless, the assumption of having cubic and spherical particles, although inspired by the SEM images, is a simplification. Surface roughness was not considered. Also stable agglomerates formed by several subparticles (e.g. Tablettose granules) are not considered in this work.

As a white reference standard for reflection Spectralon® (TheLabSphere Inc., North Sutton, USA) was used.

Spectroscopic equipment

In our experiments, we used the HyTernity-multichannel Spectrometer by Indatech. This system relies on hyperspectral pushbroom imaging which can acquire simultaneously the spectra of 30 optical fibers. The wavelength range was 900–1700 nm, which is sensitive to most pharmaceutical solid chemical products. Additionally, 3 fibers were used to directly record the intensity of the halogen light source (5 W, 12V, stabilized power supply) as a continuous reference.

For sampling the powders, the SAM-Spec® Felin probe head was used. It relies on a patented design aiming at optimal spatially resolved spectroscopy (SRS) and homogeneity measurement (full description can be found in patent WO 2010136732 A1).[57] It has a single illumination spot and different detection spots at regular distances from the illumination spot (see schematic in Figure 1). This allows to probe the distribution of light exiting the sample and to retrieve a spectra for every detection point. Simultaneously acquired spectra can be compared to quantify the spatial homogeneity of materials, or to estimate the mean path length of light inside the sample.

Ultimately, this simple (in-line) setup offers an opportunity to separate absorption from scattering. Chemometric models, which take this information into account should be more reliable when estimating concentrations and more robust with regard to variability (i.e., density and particle size variations). Successful exploitation of a related system has already been reported by Dehghani et al.,[58] showing the robustness of such a setup. Particularly, the dedicated probe

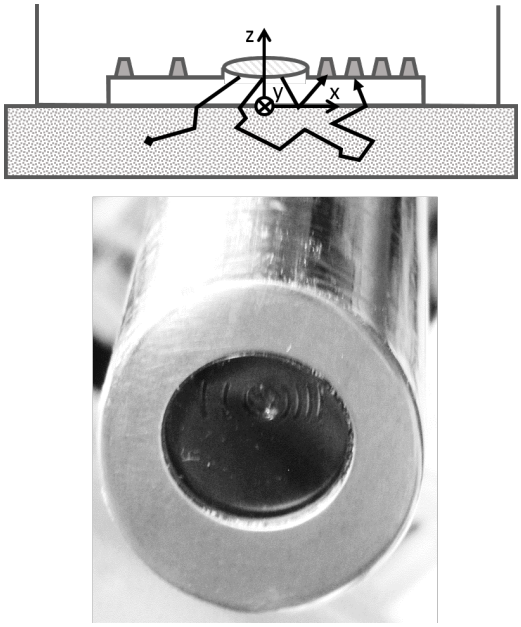


Figure 1. Schematic and picture of the Felin probe head. A central light source illuminates the powder bed. A gap exists between the light source and the powder bed, where the sapphire window is placed. Detection slits are positioned left and right. Photons are absorbed by the powder bed (left photon) or scattered and reflected inside the powder bed before being detected (two right photons). Directions and origin of the coordinate system used in the calculations is also indicated.

head was already used for the determination of API content and heterogeneity on pharmaceutical tablets by Chauchard et al.[59] and Boiret et al.[60] A similar setup, using fiber optics and a chemical imaging camera, has also been reported for tablets by Shi et al.[61]

Sample presentation

In all cases, except density variation measurements, a powder stream was poured over the tilted probe head, such that the probe was covered by the powder bed in a way that the sample could be considered optically thick (i.e., no light from external light sources reaches the detection area of the probe, and no light emitted by the probe is leaving through the back of the sample). Thus it was ensured, that the moving powder bed exceeded 5 mm thickness all the time, and left no residues on the probe head, due to sticking, etc. Furthermore, the moving powder bed should avoid subsampling effects (in this case, as all materials are the same chemically, of particle sizes and positions), as a number of spectra is integrated while particles are changing positions during flow.

Repetitions were performed by using completely new sampling and measurement. Thus, when calculating standard deviations over several repetitions, the resulting errors in intensity are quite large, although the spatial trend in reflected light intensity is persistent. This is caused by a total shift in measured intensity for all distances simultaneously in between samplings, caused by minor changes in the flow pattern and density over the probe window. As we are interested in the total intensity, simple corrective measures usually applied (namely SNV-transformation), were avoided. However, as measurements were performed over a longer period of time, spectra exhibiting the lowest and highest 10% of intensity were discarded, to filter out extremes.

The presented materials and set-up allowed to vary some relevant particle attributes. Thus, it was possible to compare materials with different particle size distributions (Granulac 70 and 140), as well as similar size distributions, but different shapes (Inhalac 120 and Granulac 70). Furthermore, density of the material was varied by increasing the pressure the probe is putting on the powder bed during measurement. The same effects were also investigated by a calculation approach.

The different grades of material available supported measurements with distinct particle attributes. Thus, it was possible to compare materials with different particle size distributions (Granulac 70 and 140), as well as similar size distributions, but different shapes (Inhalac 120 and Granulac 70). Furthermore, for some experiments, the bulk density of the material was varied by manually compressing the powder bed with the probe head during measurement. The same effects were also investigated using a calculation approach.

Calculation approach

To describe the propagation of light inside a sample of granular material, the following procedure was chosen:

1. Photons are released according to the illumination geometry of the probe above the sample.
2. For every photon:
   (a) A propagation length was determined by the geometric properties of the powder particles and voids, and material refractive index.
   (b) After a propagation step, the photon is either reflected or refracted.
   (c) When the photon leaves the sample, the photon intensity is determined by the length travelled inside material (and the resulting fraction absorbed).
3. All photons are then summed up according to their exit position, to gain the total intensity at a spatial location, to enable comparison to experimental data.

This results in an adapted MCML approach, combined with statistic assumptions about particle size and arrangement. This allows a direct connection to physical parameters, but on the other hand demands a number of simplifications. Thus the steps are discussed in detail in the following.

The model was realized in MatLab (Version 2014b, The Mathworks Inc., Natick, US).

Photon attributes

Since the photons do not carry phase information in our simulation, photon interactions and interference phenomena were neglected. Every photon was calculated separately, and the process was parallelized. Since photons plainly follow their paths determined by geometrical optics, the presented approach is only valid for $x \gg 1$.

Relying on geometrical optics is a rather crude approximation, especially since there are methods of simplified description of scattering behavior, such as the Henyey-Greenstein[62] and the Fournier-Fourand function.[63] However, these are mostly applicable to simple particle shapes. In cases of complex particle shapes and multiple scattering, the effects produced by single particles are smeared out and the inferiority of geometrical optics is irrelevant.[64,65] (For a simple proof of concept, please refer to supplementary material.)

The initial position and wavelength of the photons was chosen in order to represent the experimental light source. Their wavelength affects the probability of absorption, and direction of refraction. Furthermore, for every photon the total distance covered inside particles, as well as in voids, is stored respectively, thus allowing to calculate the intensity of photons leaving the sample.

To avoid photons being trapped inside the sample and making the calculation infinitely long, a Russian-Roulette algorithm was applied.[28]

Sample attributes

The sample particle bed contained particles of different sizes and voids. Instead of simulating the particle packing and resulting geometry, the sample was described stochastically based on its attributes, which dramatically reduced the computational effort. Furthermore, it can be argued additionally that in a moving sample the geometry is not preserved over longer measurement periods.

The sample was located in the half space $z < 0$, (as shown in Figure 1). The photons are emitted at $z = 0 + \varepsilon$, with ε being small. Thus, when the photon reaches a position $z > 0$ again, it must have exited the sample and can be detected.

A photon traveling inside the sample alternates between traveling in free space and in the particle material, separated at the interfaces where refraction can occur. The orientation of the interfaces was assumed uniformly random.

Photon-interface behavior

At an interface, the photons obey the laws of geometrical optics. Reflection, total reflection and refraction occur according to Snells law. Whether transmission or reflection takes place is determined by reflection and transmission intensity according to the Fresnel laws, where the relative reflection and transmission (compared with the incoming intensity) determine the possibility of the respective behavior.

Wavelength dependent refractive index

The refractive index of the material depends on the wavelength. Since there are few measurements of refractive indices of lactose as solid (and not in solution) in the required wavelength range,[66] the Sellmeier dispersion model of the refractive index was applied:

$$n_{\text{material}} = \sqrt{1 + \frac{B_1 \cdot \lambda^2}{\lambda^2 - C_1}} \tag{1}$$

with constants $B_1 = 1.124$ and $C_1 = 0.01187\ \mu\text{m}^2$. Those coefficients were determined by a least squares fit from experimental data[67] (more details are in the supplemental material). The orientation dependence of the refractive index was neglected. The refractive index of the voids (or air) was set to $n_{\text{void}} = 1.0$ (constant for all wavelengths).

Photon step length

The step length, i.e., the distance between the interfaces, is different for photons inside material particles and inside voids between material particles.

Path length inside material particles

The possibility of finding a certain step length inside and between the borders of a particle is given by the chord length distribution. It can be calculated for known particle shapes and simple geometries.[68,69] Generally, the inverse problem (i.e., determining the particle size by the chord length) is of interest in measurement applications.[70,71] However, the possible path length of a light ray passing through the particle is different from the chord length distribution: refraction occurs when light enters the particle and the total reflection inside the particle may lengthen the path.

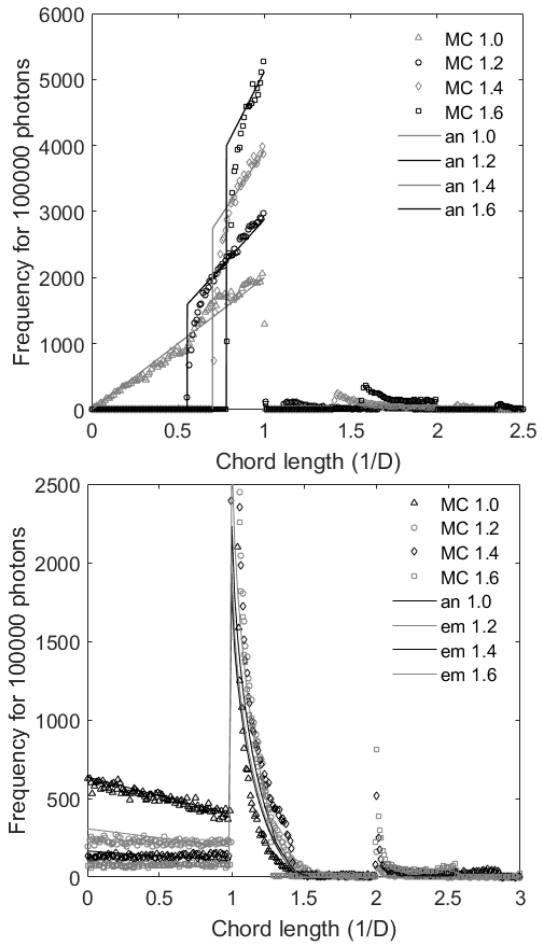


Figure 2. Chord length distribution for various refraction numbers obtained via MC simulations with 100,000 test photons (MC), analytic solutions (an) and empirical functions (em). The numbers indicate the refractive index. Top: Distribution for a sphere with diameter D. Bottom: Distribution for a cube with side length D.

To include the entry refraction, Monte-Carlo ray-tracing calculations for single particles were performed, with incoming rays distributed μ-uniformly. These calculations were repeated for various refractive numbers n_{material} and the results were compared with known analytical solutions for chord length distributions. In the case of $n_{\text{material}} = 1.0$, (i.e., no refraction at the interface), the Monte-Carlo results and the analytical solutions agreed. A useful overview of analytical chord length distributions for various shapes was collected by Bradley.[72] For particles with higher refractive indexes, the chord length distribution changes its shape, as shown for two geometries in Figure 2.

The analytical formulas were adapted to reflect changes in the chord length distribution for various refractive indexes. Hence, for spheres with diameter D and refractive index n the resulting distribution q_D of chord lengths s is given by the analytical description (see supplementary material for derivation)

$$q_D(s) = \begin{cases} 2\frac{n^2}{D^2}s & , \sqrt{\frac{n^2-1}{n^2}} < s < D \\ 0 & , \text{otherwise} \end{cases} \quad (2)$$

For cubes with side length D and refractive index n, a semi-empirical distribution is:

$$q_D(s) = \begin{cases} \frac{8}{3D\pi}\left(\frac{1}{n^4}\right) & ;\xi = 0 \\ k\left(8\xi^3 - 3\xi^4\right)\left(\frac{1}{n^4}\right) & ;0 < \xi \leq 1 \\ k\left(6\pi + 6\xi^4 - 1 - 8\left(2\xi^2+1\right)\sqrt{\xi^2-1}\right)\left(\frac{n^2+1}{2n^2}\right) & ;1 < \xi \leq 1 + \frac{\sqrt{2}-1}{n} \\ k\left(6\pi - 3\xi^4 - 5 + 8\left(\xi^2+1\right)\sqrt{\xi^2-2} - 24\text{atan}\left(\sqrt{\xi^2-2}\right)\right)\left(\frac{n^2+1}{2n^2}\right) & ;1 + \frac{\sqrt{2}-1}{n} < \xi \leq 1 + \frac{\sqrt{3}-1}{n} \\ 0 & ;1 + \frac{\sqrt{3}-1}{n} < \xi \leq \sqrt{3} \\ & \text{with } \xi = \frac{s}{D} \text{ and } k = \frac{1}{3D\pi\xi^3} \end{cases} \tag{3}$$

Both distributions are consecutively normalized, i.e., $\hat{q}_D(s) = \frac{q_D(s)}{\int_0^\infty q_D(s) \quad ds}$.

Overall, including refraction increases the probability of longer chord lengths because convex particles must have positive curvature, and thus refract incoming rays towards the center of the particle, where longer path lengths dominate. The longest possible lengths are $1 \cdot D$ for the sphere and $\sqrt{3} \cdot D$ for the cube. Furthermore, total reflections of rays leaving the object lead to a reappearance of chord length probability with higher lengths. Even higher orders exist, with increasingly lower probability, as also shown in Figure 2.

The diameter D of individual particles is not fixed and follows a distribution. Here, we assumed a log-normal distribution to be:

$$l(D) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{\left(\log(D) - \log(D_{50})\right)^2}{2\log^2(\sigma)}\right) \tag{4}$$

$$\sigma \approx \frac{D_{84}}{D_{16}} \approx \frac{D_{90}}{D_{50}} \approx \frac{D_{50}}{D_{10}} \tag{5}$$

where $D_{\%}$ are the corresponding particle fraction sizes. For every (also empirical) particle size distribution and known chord size distribution related to the individual particle shape, the chord size distribution of the particle collective can be calculated by

$$p_c(s) = \frac{\int_0^\infty \hat{q}_D(s) l(D) D \quad dD}{\int_0^\infty l(D) D \quad dD} \tag{6}$$

Path lengths in voids between material particles

The length and direction of rays that pass through voids between particles and the resulting chord lengths depend on the packing and porosity of the particle bulk. It is often assumed that the chord distribution in voids follows an exponential law. This is true if spheres are randomly positioned, resulting in the Poisson sphere distance distribution. Since the particle frequency varies with porosity, a commonly-used approximation for the void chord lengths in a material with a typical particle diameter D and porosity φ is [73,74]

$$q(s) = \frac{1}{\mu}\exp(-\mu s)\,; \text{ with } \mu(\varphi, D) = \frac{3\varphi D}{2(1-\varphi)} \tag{7}$$

Although this function also applies to tightly packed spheres, it is not true for small distances. In practice, the distribution of small distances is [75]

$$q(s) \propto n_c\sqrt{s} \tag{8}$$

where n_c is the typical number of contact points between the particles. This behavior can be approximated via a linear function when the turning point (which is a function of porosity) leading to the Poisson behavior is known. [49,76]

In this work, we assumed the turning point to be located at D/φ, resulting in the following chord length distribution in the voids, (after normalization):

$$q(s) = \begin{cases} \frac{3\varphi^2}{D^2\left(\frac{7}{2}-2\varphi\right)} s \;;\; 0 < s \leq \frac{D}{\varphi} \\ \frac{3\varphi}{D}\left(\frac{1}{\frac{7}{2}-2\varphi}\right)\exp\left(\frac{3}{2}\left(\frac{1}{1-\varphi}\right)\left(1 - \frac{\varphi}{D}s\right)\right) \;;\; \frac{D}{\varphi} < s \end{cases} \tag{9}$$

For the purpose of calculating the void chord length distribution, the broadness of particle size distribution and the shape of particles were neglected.

Photon absorption

While travelling inside a material, photons can be absorbed, and the probability of absorption depends on the wavelength. Absorption probability $\epsilon(\lambda)$ is modeled by a sum of i Gaussian peaks, centered around the known absorption features at wavelengths $\hat{\lambda}_i$. A more detailed description and the necessary parameters can be found in the supplementary material.

The total path length of a photon travelling inside a material was summed up and the probability for absorption p was calculated. Assuming a photon leaving the sample has travelled for a length of l inside material, the absorbed photon intensity (i.e., the probability of absorption for every single photon) is given by the specific absorptivity and the path length via the well-known Beer-Lambert law [77]

$$p(\epsilon, l) = 1 - \exp(-f\epsilon(\lambda)\, l) \tag{10}$$

with a factor f, which is required since the specific absorptivity was determined using a setup with an unknown path length (an attribute of any diffuse reflection spectroscopy). f links the absorption probability to a certain length scale and is related to the scattering and absorption ratios and mean free path lengths of photons. To choose a reasonable relative absorption a viable approximation is $f = 1\ \mu\text{m}^{-1}$. Varying f leads to a shift of finally estimated absorbance. It is known that, generally, in the near infrared region scattering and absorption coefficients differ by one order of magnitude,[17,78] which is also reproduced here.

Photon detection

When photons left the sample, their point of exit was recorded. Summing up all photons leaving the sample should give the intensity distribution of light on the samples surface. A radial symmetry exists. The photons are summed up and normalized (per area) on annuli with the increasing radii to illustrate the intensity decrease in relation to the distance from the illumination spot.

Comparing calculated photon count to experimentally determined intensity and absorbance

Some properties of the experimental setup must be included in the calculations to make the results comparable. Mainly these are the wavelength dependent illumination intensity and detection sensitivity. Furthermore optical elements that result in (wavelength dependent) absorption or beam shape change (like lenses) must be considered.

In a typical NIR spectrometer setup, all these influences are assessed by measuring a white standard, which should be positioned in the expected sample position and be totally diffuse reflecting in the corresponding wavelength range. The absorbance (generally, the chemometric variable of interest) is calculated as $A = -\log(I/I_w)$, where I and I_w are the measured intensity of the sample and white reference, respectively.

Instead of recreating the detailed probe geometry in simulation (compare to Figure 1), the initial spatial photon distribution was assumed to follow a diffuse light source with radius a at distance d, which is given analytically by[79]

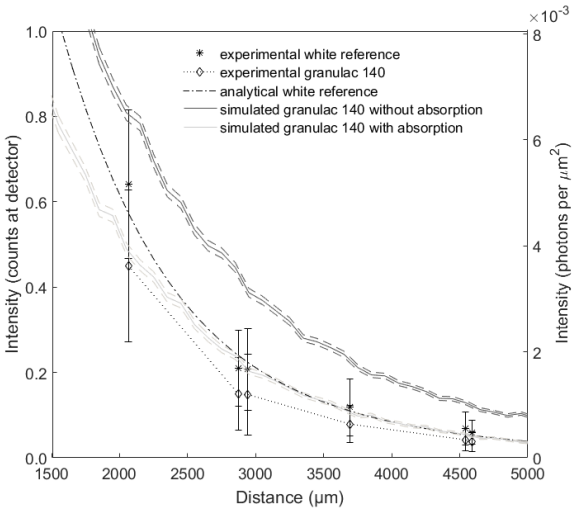


Figure 3. Intensity drop with distance of illumination. The analytical function as denoted in Equation 11 is shown, which is in agreement with the measurement of the white reference. Furthermore the simulation results for intensity for Granulac 140 are depicted (together with uncertainty estimated), with and without absorption. Scattering at the particles broadens the initial (analytical) distribution. Absorption decreases the intensity notably and cannot be neglected in simulation. Simulated and also shown experimental results for Granulac 140 are in agreement.

$$I_z(r) = I_0 \frac{\pi}{2} \left(1 - \frac{r^2 + d^2 - a^2}{\sqrt{r^4 + 2\,(d^2 - a^2)\, r^2 + (d^2 + a^2)^2}} \right) \tag{11}$$

$$\text{except } d = 0 \text{ for } r < a \tag{12}$$

This solution is isotropic on the plane $z = 0$, r is the distance from the origin, $a = 300\ \mu$m describes the radius of the illumination source (i.e., a 600 μm diameter fiber) and $d = 1.75 \cdot 1$ mm is the effective optical thickness of the sapphire window, assuming a refractive index for sapphire of 1.75 across all wavelengths.[80] Additionally the probability of photons having a certain wavelength $p_\lambda(\lambda)$ follows the experimentally-detected intensity.

This illumination distribution is used for two cases, first as probability for the initial photon distribution, and second as illumination density for the white standard in the MC approach.

The experimentally measured white reference, and the analytically approximated can now be compared, as shown in Figure 3, to establish a fixed factor between the number of simulated photons, and the number of photons detected on the chip in the experimental setup. This results in an irradiance of $I_0 = 1.238\ \frac{\text{photons}}{\mu\text{m}^2\text{s}}$, which is constant as long as the experimental setup does not change.

For the purposes of calculations, not only the absolute number of photons should be established but a variable similar to absorbance should be chosen that does not depend on the number of photons emitted from the light source. This can now be done effectively by the relation

$$A\left(r,\lambda\right) = -\log\left(\frac{I_{\mathrm{MC}}\left(r,\lambda\right) I_0}{p_\lambda\left(\lambda\right) N I_z\left(r\right)}\right) \tag{13}$$

i.e., the number of simulated photons $I_{\mathrm{MC}}(r,\lambda)$ detected at distance r, is compared to the number of photons expected $I_z(r)$, and normalized by the number of emitted photons N and the probability of having wavelength λ. This way, the calculated absorbance is independent of the number of photons emitted, although a higher number naturally results in less statistical noise.[81,82]

Summary

The parameters needed for the simulation are summarized in Table 2. It is noteworthy, that none of the parameters is a fudge parameter used to optimize the simulation. All parameters are provided by external measurements on the sample, or the spectrometer. The only parameter used to bring into agreement simulation and experimental data is the irradiance of simulated photons (i.e. a single factor adjusting the simulated light intensity, in order to achieve the same number of counts in simulation and on the experimental detector chip), which is determined by measuring a white standard material, and compared to Equation 11.

It can be seen, that thus simple information about the particles can be used to assess their influence on the recorded spectra. Furthermore, wavelength dependencies are included in the calculations, which are often omitted in regular Monte-Carlo light transport applications.

Results

Using the materials listed in Table 1, we were able to selectively vary the particle attributes. Most variations were reflected in a change in the relationship between the detected intensity (summed over all wavelengths to improve the statistics) and the distance to the illumination point.

Variations in particle size, shape and bulk density

We used the chemically identical substance and varied the particle size, expecting different scattering results. Model substances were Granulac 140 and Granulac 70, with a relative mean particle size ratio of 2:3. The numerical and experimental results can be found in Figure 4. The simulated

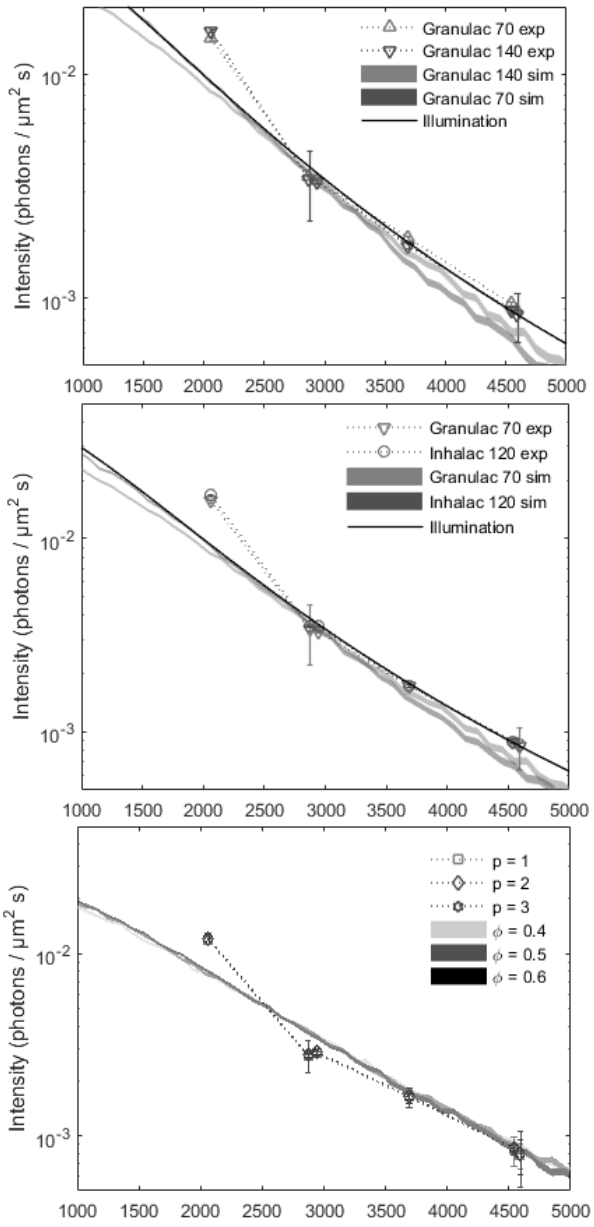


Figure 4. Variation of intensity with varying particle parameters. The single points represent experimental measurements and the error bars indicate standard deviations by repeated sampling and measurement. The bands indicate the Monte-Carlo calculations with plus/minus one standard deviation (with 200,000 photons). Top: Intensity for different particle sizes. Center: Different particle shapes. Bottom: Different powder porosity values.

results produced the same shape for the intensity drop, but exhibited a larger difference in intensity between those two substances than seen experimentally.

Variations in the particle shape were observed when comparing Granulac and Inhalac particles. Crudely approximated, Granulac particles are spherical and Inhalac particles are cube-shaped. To ensure roughly the same particle sizes, Granulac 70 was compared to Inhalac 120. Please note, however, that there still were some differences in the particle

Table 2. Parameters fed to the simulation. Those marked with "material dep." are determined per material and are chosen accordingly, as listed in Table 1.

Item	Value	Comment
spectrometer/probe dependent		
illuminating disk diameter	600 µm	optical fiber diameter
distance to powder bed	1.75 mm	optical thickness of 1 mm sapphire window
energy distribution	prob. dist.	determined experimentally via white standard
luminosity of light source	$1.238\ \frac{\text{photons}}{\mu\text{m}^2\text{s}}$	summary parameter of light source intensity, detector chip sensitivity and illumination time
absorptivity factor	$1\ \frac{1}{\mu\text{m}}$	ratio of absorbance reference to simulation
material dependent		
particle size	material dep.	mean particle diameter D_{50}
particle size width	material dep.	calculated from particle size distribution (as in Equation 5)
particle shape assumption	spherical/cubic	assumption for simplification
refractive index	Sellmeier equation	this is explained in detail in supplementary material
porosity	material dep.	experimentally determined by bulk and true density
refractive index of air	1.0	assumed constant

size, as listed in Table 1. The change in the intensity drop-off is shown in Figure 4. Compared to the previous size comparison, the change in intensity is greatly reduced. Only close to the illumination spot, the cubic particles exhibit higher intensity, compared to the spherical ones. Here, MC calculations and experiment are in agreement, in showing only a slight difference (as compared to particle size).

Variations in the pressure applied to the probe head immersed in the powder bed inevitably lead to a change in porosity in front of the probe head. The change in the resulting intensity pattern is shown in Figure 4. Unfortunately, since the probe head was operated manually, the actual density of the powder is not known. Nonetheless, calculations for various porosity levels can be found in the same Figure 4. Notably, experiments and simulations show the same behavior, i.e. the difference in intensity in between porosity levels is smaller than the uncertainty for a single measurement or simulation.

Absorbance spectra and sampled volume

Naturally, effective path length and thus absorbance increases with distance of detection to illumination spot.

Figure 5 shows spectra according to the experimental results as well as computationally acquired spectra. The numerical results agree with the experimental data, in terms of relative increase of absorbance with distance. To achieve the same absolute scale, the factor f (as in Equation 10) could be adjusted.

Not only do the different detection spots correlate to different distance of the illumination source, but every spot actually represents a different sampled volume. The MC calculations now allow to estimate the shape and size of these volumes. This otherwise unavailable information now can go beyond the qualitative statement, that the surface dominates the spectra, but can show, how changes in the sample, e.g. the particle size, change also the sampled volume.

The position of scattering events was recorded, allowing to locate the parts of the sample volume exhibiting increased interaction. The resulting sampled volume and the depth profile are shown in Figure 6, for two different particle sizes.

Discussion

The obtained results can be interpreted in terms of the influences of different particle properties.

It is evident that larger particles have larger voids between particles. A direct result is that larger particles have a greater penetration depth of light. This leads to a less distinct radial decrease of intensity from the point of illumination. As smaller particles confine the photons to a smaller volume, there is a more intense reflection near the illumination point and thus less intensity found on the sample surface at more remote locations, resulting in a sharper decrease of intensity. It thus depends on the measurement position, if the obtained intensity coming from the sample is stronger or weaker than the illumination intensity. These general tendencies were also observed in the experimentally acquired spectra. Of course, polydisperse and polymodal particle size distributions are more complex and need to be studied more carefully.

Interestingly, the different shapes did not lead to such a sharp distinction, neither experimentally nor in simulation. This supports the argument, that individual particle characteristics are suppressed to a large degree in the bulk collective.

Leaving the particle properties identical, but changing the bulk porosity, and hence packing density, naturally effects the penetration of light into the powder bed. However, there was no significant change in the intensity distribution, neither

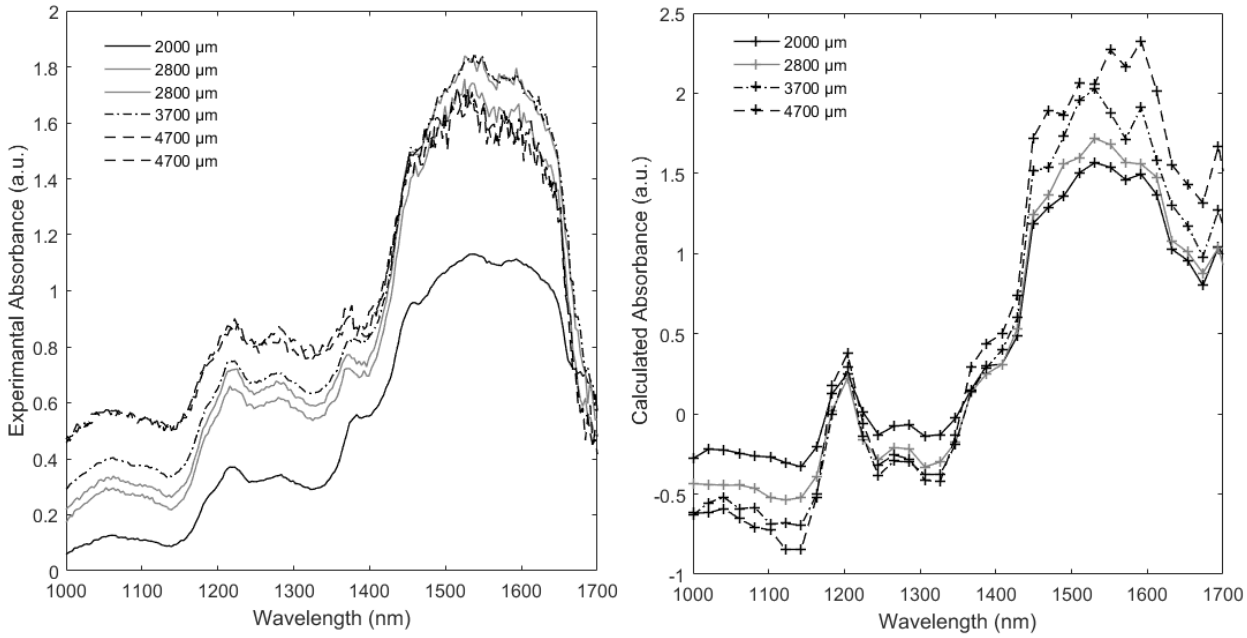


Figure 5. Experimentally determined (left) and calculated absorbance spectra (right).

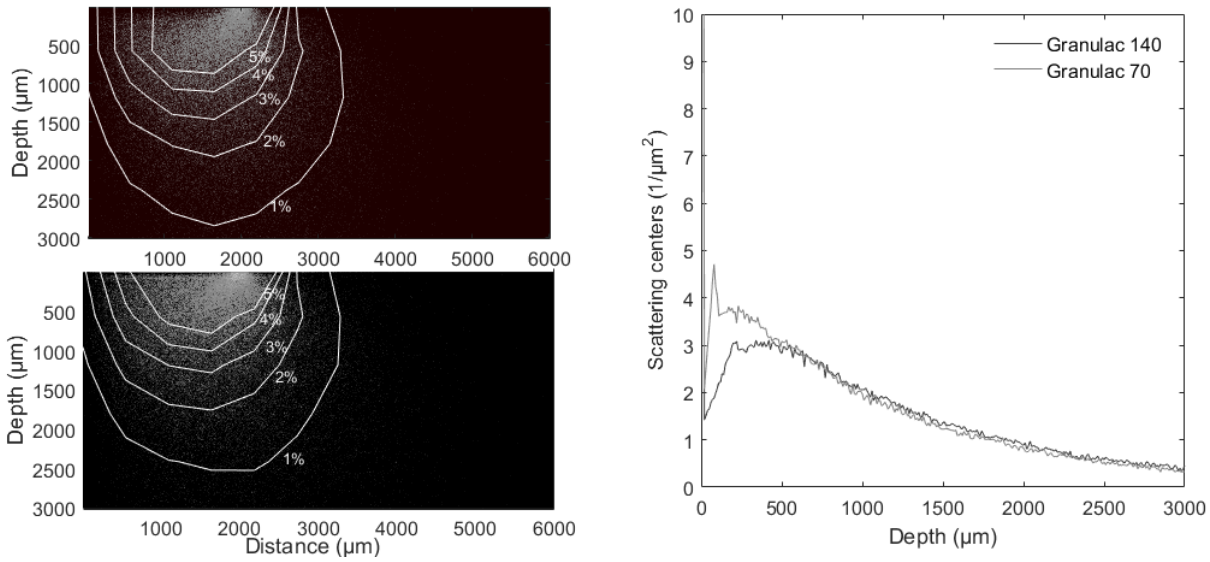


Figure 6. Left: Two dimensional representation of the sampled volume for a detector located at 1.9 to 2.1 mm distance, for Granulac 140 (top) and Granulac 70 (bottom). The lines indicate regions where 0.25 mm^3 contribute 1, 2, 3, 4, 5% to the total detected intensity respectively. Right: Direct comparison of the depth profile contributing to the measurement of the two different powders.

experimental nor in simulation. Thus the sampled volume and the sampled mass do not go hand in hand.

Yet still open is the important issue of the sample volume that contributes to the detected intensity. In Figure 6 only photons that reached a certain detection area are shown. The sample volume is rather shallow and has no sharp border but rather an exponential decrease towards the surrounding mass. It can be shown that the penetration depth, and the sampled volume, increases with increasing sampling distance from the illumination, thus clearly showing a benefit of spatially offset spectroscopy.

Additionally, it can be shown in which way the sampled volume changes with changing particle properties. Smaller particles lead to a narrower sampled volume. This can help explaining and counteracting non-linearity in concentration measurements of blends of different powders.

Clearly missing from the calculations though is the scattering effect caused by the fine particle fraction, which leads to the curved background as also indicated in Figure 5. Surface roughness and scattering at particles with $x \approx 1$ was not taken into account, and seems necessary to achieve this effect. Although the dependence of the refractive

index on the wavelength is considered via the Sellmeier approximation, this is not sufficient in replicating the scattering wavelength dependence.

Nonetheless, the presented work confirms that the chord length distributions carry enough information to predict the behavior of different particle size and shape properties, and the actual particle positions need not to be considered in the MC simulation.

Summary and outlook

Understanding the propagation of light in pharmaceutical powder blends can help to estimate, first, the sample volume represented in the spectra and, second, the effective path length of light inside the sample. This allows decoupling of absorption and scattering and offers a more robust interpretation of the acquired spectra for the determination of concentrations.

Thus, we studied several properties that influence light diffusion in granular media experimentally using the Indatech SAM-Spec® Felin probe head for spatial sampling of the specimen, as well as numerically via a modified Monte-Carlo approach. This demanded several simplifications, but it was possible to copy the expected behavior and demonstrate how particle size, shape and bulk density affect the detected light intensity, by using those as direct input parameter. The simulations clearly facilitate the interpretation of the acquired spectra.

Spectra acquired using a spatially resolved probe head (e.g., the Felin head) can effectively be used to extract information about the powder bed in an elaborate manner. In a next step, the reverse could be of interest, i.e. determining particle size and sample volume from the spectra. Finally, it would be worth to have a method that based on the spatially resolved spectra would allow to give not only relative concentrations, but also absolute masses by including the information on the sample volume and its change.

Although the proposed simulations are rather simple, they are efficient and rather fast. In the future, other properties (e.g., particle roughness and scattering on smaller particles) can be included to approximate the response of a real powder even better. The method is straightforward and can be expanded to blends of various materials, with their own particle sizes, shapes and absorption characteristics. It will be particularly interesting to apply it to materials with a blend of chemical and physical characteristics (e.g., various pharmaceutical excipients and APIs).

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Supplemental material

Some details which would unnecessary lengthen the text, but form the basis for the presented work, are found in the supplementary material. Specifically, these topics are:

- Particle size distributions of the investigated materials
- Fitting the Sellmeier dispersion equation
- A priori chord placement probabilities
- Derivation of chord length distribution in a refracting sphere
- Comparison of chord length distributions in spherical and cubic particles and in voids
- Comparison of geometric and accurate scattering on single particles
- Comparison of object-based and chord-length-based scattering
- Absorption characteristics of α-lactose monohydrate
- Photon energy distributions in the experimental and simulated setup

References

1. U.S. Food and Drug Administration. *Guidance for Industry: PAT - A Framework for Innovative Pharmaceutical Development, Manufacturing and Quality Assurance*, 2004.
2. K. A. Bakeev (editor.) *Process Analytical Technology: Spectroscopic Tools and Implementation Strategies for the Chemical and Pharmaceutical Industries*. 2 ed. Oxford: Wiley, 2010. ISBN 978-0-470-72207-7. DOI:10.1002/9780470988459.
3. C. P. O'Donnell, C. Fagan and P. J. Cullen. *Process Analytical Technology for the Food Industry*. New York: Springer Science+Business Media, 2014. ISBN 9780123744883. DOI: 10.1016/S1571-5078(08)00422-4.
4. W. Chew and P. Sharratt. "Trends in Process Analytical Technology". *Anal Methods*. 2010. 2(10): 1412. DOI:10.1039/c0ay00257g.
5. L. L. Simon, H. Pataki, G. Marosi et al. "Assessment of Recent Process Analytical Technology (PAT) Trends: A Multiauthor Review". *Org Process Res Dev*. 2015. 19(1): 3–62. DOI: 10.1021/op500261y.

6. M. Jamrógiewicz. "Application of the Near-infrared Spectroscopy in the Pharmaceutical Technology". *J Pharm Biomed Anal*. 2012. 66: 1–10. DOI:10.1016/j.jpba.2012.03.009.
7. Y. Ozaki. "Near-infrared Spectroscopy - its Versatility in Analytical Chemistry". *Anal Sci*. 2012. 28(6): 545–63. DOI:10.2116/analsci.28.545.
8. O. Scheibelhofer, D. M. Koller, P. Kerschhaggl et al. "Continuous Powder Flow Monitoring via Near-infrared Hyperspectral Imaging". In *2012 IEEE International Instrumentation and Measurement Technology Conference Proceedings*. IEEE. ISBN 978-1-4577-1772-7, 2012. pp. 748–753. DOI:10.1109/I2MTC.2012.6229431.
9. R. W. Kessler. "Perspectives in process analysis". *J Chemom*. 2013. 27(11): 369–378. DOI:10.1002/cem.2549.
10. O. Scheibelhofer, N. Balak, D. Koller et al. "Spatially Resolved Monitoring of Powder Mixing Processes via Multiple NIR-probes". *Powder Technol*. 2013. 243: 161–170. DOI:10.1016/j.powtec.2013.03.035.
11. F. Westad, A. Schmidt and M. Kermit. "Incorporating Chemical Band-Assignment in Near Infrared Spectroscopy Regression Models". *J Near Infrared Spectrosc*. 2008. 16(3): 265–273. DOI:10.1255/jnirs.786.
12. E. W. Ciurczak and B. Igne. *Pharmaceutical and Medical Applications of Near-Infrared Spectroscopy*. 2nd ed. Boca Raton: CRC Press, 2015. ISBN 978-1-4987-0446-5.
13. M. Pasikatan, J. Steele, C. Spillman et al. "Near infrared reflectance spectroscopy for online particle size analysis of powders and ground materials". *J Near Infrared Spectrosc*. 2001. 9(1): 153. DOI:https://doi.org/10.1255/jnirs.303.
14. M. Otsuka, Y. Mouri and Y. Matsuda. "Chemometric Evaluation of Pharmaceutical Properties of Antipyrine Granules by Near-infrared Spectroscopy". *AAPS PharmSciTech*. 2003. 4(3): E47. DOI:10.1208/pt040347.
15. O. Scheibelhofer, B. Grabner, R. W. Bondi et al. "Designed Blending for Near Infrared Calibration". *J Pharm Sci*. 2015. 104(7): 2312–2322. DOI:10.1002/jps.24488.
16. D. J. Dahm and K. D. Dahm. "The Physics of Near Infrared Scattering". In P. Williams and K. Norris (editor.) *Near Infrared Technology in the Agricultural and Food Industry*, 2 ed, chapter 1. St. Paul: American Association of Cereal Chemists, 2004. pp. 1–17.
17. W. Kessler, D. Oelkrug and R. Kessler. "Using Scattering and Absorption Spectra as MCR-hard Model Constraints for Diffuse Reflectance Measurements of Tablets.". *Anal Chim Acta*. 2009. 642(1-2): 127–34. DOI:10.1016/j.aca.2009.01.057.
18. Z. Shi and C. A. Anderson. "Pharmaceutical Applications of Separation of Absorption and Scattering in Near-Infrared Spectroscopy (NIRS)". *J Pharm Sci*. 2010. 99(12): 4766–4783. DOI:10.1002/jps.
19. D. L. Wetzel. "Near-Infrared Reflectance Analysis". *Anal Chem*. 1983. 55(12): 1165–1176. DOI:10.1021/ac00262a718.
20. A. A. Kokhanovsky (editor.) *Light Scattering Reviews - Single and Multiple Light Scattering*. Chichester, United Kingdom: Praxis Publishing, 2006. ISBN 978-3-540-25315-0. DOI:10.1007/3-540-37672-0.
21. G. G. Poludniowski and P. M. Evans. "Optical Photon Transport in Powdered-Phosphor Scintillators. Part 1. Multiple-scattering and Validity of the Boltzmann Transport Equation". *Med Phys*. 2013. 40(4): 041904. DOI:10.1118/1.4794483.
22. A. Hohmann, F. Voit, J. Schäfer et al. "Comparison of monte carlo simulations with exact maxwell solutions for polarized light scattering by multiple absorbing spheres". *J Phys Conf Ser*. 2012. 369: 012007. DOI:10.1088/1742-6596/369/1/012007.
23. C. F. Bohren and D. R. Huffman. *Absorption and Scattering of Light by Small Particles*. New York: Wiley-VCH Verlag GmbH & Co KG, 2004.
24. Y. L. Xu and B. Å. S. Gustafson. "A Generalized Multiparticle Mie-solution: Further Experimental Verification". *J Quant Spectrosc Radiat Transfer*. 2001. 70(4-6): 395–419. DOI:10.1016/S0022-4073(01)00019-X.
25. S. L. Jacques. "Optical Properties of Biological Tissues: A Review". *Phys Med Biol*. 2013. 58(11): R37–61. DOI:10.1088/0031-9155/58/11/R37.
26. M. Kervella, F.-X. D'Abzac, F. Hache et al. "Picosecond Time Scale Modification of Forward Scattered Light Induced by Absorption Inside Particles". *Opt Express*. 2012. 20(1): 32–41. DOI:10.1364/OE.20.000032.
27. J. D. Rogers, A. J. Radosevich, J. Yi et al. "Modeling Light Scattering in Tissue as Continuous Random Media Using a Versatile Refractive Index Correlation Function". *IEEE J Sel Topics in Quantum Electron*. 2014. 20(2): 7000514. DOI:10.1109/JSTQE.2013.2280999.
28. L. Wang, S. L. Jacques and L. Zheng. "MCML - Monte Carlo Modeling of Light Transport in Multi-layered Tissues". *Computer Methods and Programs in Biomedicine*. 1995. 47: 131–146. DOI:10.1016/0169-2607(95)01640-f.
29. D. K. McAllister, A. Lastra and W. Heidrich. "Efficient Rendering of Spatial Bi-directional Reflectance Distribution Functions". In *Proceedings of the ACM SIGGRAPH/EUROGRAPHICS conference on Graphics hardware*. Saarbrücken: Eurographics Association, 2002. pp. 79–89.
30. Y. M. Govaerts and M. M. Verstraete. "Raytran: A Monte Carlo Ray-tracing Model to Compute Light Scattering in Three-dimensional Heterogeneous Media". *IEEE Trans Geosci Remote Sens*. 1998. 36(2): 493–505. DOI:10.1109/36.662732.

31. T. Ritschel, C. Dachsbacher, T. Grosch et al. "The State of the Art in Interactive Global Illumination". *Comput Graphics Forum*. 2012. 31(1): 160–188. DOI:10.1111/j.1467-8659.2012.02093.x.
32. G. H. Goedecke. "Radiative Transfer in Closely Packed Media". *J Opt Soc Am*. 1977. 67(10): 1339–1348. DOI:10.1364/JOSA.67.001339.
33. P. Kubelka. "New Contributions to the Optics of Intensely Light-Scattering Materials Part I". *J Opt Soc Am*. 1948. 38(5): 448. DOI:10.1364/josa.38.000448.
34. Y. Zhang and G. McGeorge. "Quantitative Analysis of Pharmaceutical Bilayer Tablets Using Transmission Raman Spectroscopy". *J Pharm Innovation*. 2015. 10(3): 269–280. DOI:10.1007/s12247-015-9223-8.
35. O. Berntsson, L.-G. Danielsson and S. Folestad. "Estimation of Effective Sample Size when Analysing Powders with Diffuse Reflectance Near-Infrared Spectrometry". *Anal Chim Acta*. 1998. 364(1-3): 243–251. DOI:10.1016/s0003-2670(98)00196-2.
36. Y. Roggo, P. Chalus, L. Maurer et al. "A Review of Near Infrared Spectroscopy and Chemometrics in Pharmaceutical Technologies". *J Pharm Biomed Anal*. 2007. 44(3): 683–700. DOI:10.1016/j.jpba.2007.03.023.
37. S. C. Kanick, D. J. Robinson, H. J. C. M. Sterenborg et al. "Monte Carlo Analysis of Single Fiber Reflectance Spectroscopy: Photon Path Length and Sampling Depth.". *Phys Med Biol*. 2009. 54(22): 6991–7008. DOI:10.1088/0031-9155/54/22/016.
38. T. Papaioannou, N. W. Preyer, Q. Fang et al. "Effects of Fiber-optic Probe Design and Probe-to-target Distance on Diffuse Reflectance Measurements of Turbid Media: An Experimental and Computational Study at 337 nm.". *Appl Opt*. 2004. 43(14): 2846–2860. DOI:10.1364/AO.43.002846.
39. A. Arhaliass and P. Snabre. "Multiple light scattering in random systems: Analysis of the backscattering spot image". *Eur Phys J AP*. 1999. 6(1): 81–87. DOI:10.1051/epjap:1999155.
40. Z. Shi and C. A. Anderson. "Application of Monte Carlo Simulation-Based Photon Migration for Enhanced Understanding of Near-Infrared (NIR) Diffuse Reflectance. Part I: Depth of Penetration in Pharmaceutical Materials". *J Pharm Sci*. 2010. 99(5): 2399–2412. DOI:10.1002/jps.22013.
41. C. Abrahamsson, J. Johansson, S. Andersson-Engels et al. "Time-Resolved NIR Spectroscopy for Quantitative Analysis of Intact Pharmaceutical Tablets". *Anal Chem*. 2005. 77(4): 1055–1059. DOI:10.1021/ac0487754.
42. C. Pilorget, M. Vincendon and F. Poulet. "A Radiative Transfer Model to Simulate Light Scattering in a Compact Granular Medium using a Monte-Carlo Approach: Validation and First Applications". *J Geophys Res: Planets*. 2013. 118(12): 2488–2501. DOI:10.1002/2013JE004465.
43. T. Okamoto, T. Kumagawa, M. Motoda et al. "Monte Carlo Simulation of Light Reflection from Cosmetic Powder Particles near the Human Skin Surface". *J Biomed Opt*. 2013. 18(6): 61232. DOI:10.1117/1.JBO.18.6.061232.
44. N. T. Tran, C. G. Campbell and F. G. Shi. "Study of Particle Size Effects on an Optical Fiber Sensor Response Examined with Monte Carlo Simulation.". *Appl Opt*. 2006. 45(29): 7557–7566. DOI:10.1364/AO.45.007557.
45. Z. Sadjadi and M. Miri. "Diffusion of Light in Two-dimensional Granular Materials". *AIP Conference Proceedings*. 2013. 1542: 109–112. DOI:10.1063/1.4811879. arXiv:1203.3938v1.
46. T. Svensson, K. Vynck, M. Grisi et al. "Holey Random Walks: Optics of Heterogeneous Turbid Composites". *Phys Rev E: Stat, Nonlinear, Soft Matter Phys*. 2013. 87(2): 022120. DOI:10.1103/PhysRevE.87.022120. arXiv:1211.1954v1.
47. T. Svensson, K. Vynck, E. Adolfsson et al. "Light diffusion in quenched disorder: Role of step correlations". *Phys Rev E*. 2014. 89(2). DOI:10.1103/PhysRevE.89.022141.
48. A. B. Kostinski. "On the Extinction of Radiation by a Homogeneous but Spatially Correlated Random Medium". *J Opt Soc Am A*. 2001. 198(8): 1929–1933. DOI:10.1364/JOSAA.19.002517.
49. G. L. Olson, D. S. Miller, E. W. Larsen et al. "Chord Length Distributions in Binary Stochastic Media in Two and Three Dimensions". *J Quant Spectrosc Radiat Transfer*. 2006. 101(2): 269–283. DOI:10.1016/j.jqsrt.2005.11.070.
50. J. M. Dlugach, M. I. Mishchenko, L. Liu et al. "Numerically Exact Computer Simulations of Light Scattering by Densely Packed, Random Particulate Media". *J Quant Spectrosc Radiat Transfer*. 2011. 112(13): 2068–2078. DOI:10.1016/j.jqsrt.2011.02.009.
51. V. P. Tishkovets, E. V. Petrova and M. I. Mishchenko. "Scattering of Electromagnetic Waves by Ensembles of Particles and Discrete Random Media". *J Quant Spectrosc Radiat Transfer*. 2011. 112(13): 2095–2127. DOI:10.1016/j.jqsrt.2011.04.010.
52. E. Alerstam, T. Svensson, S. Andersson-Engels et al. "Single-fiber diffuse optical time-of-flight spectroscopy". *Opt Lett*. 2012. 37(14): 2877–2879. DOI:10.1364/OL.37.002877.
53. A. Kienle and M. S. Patterson. "Determination of The Optical Properties of Turbid Media From a Single Monte Carlo Simulation.". *Phys Med Biol*. 1996. 41(10): 2221–2227.
54. Xiang Wen, Xiewei Zhong, Tingting Yu et al. "A Monte Carlo Based Lookup Table for Spectrum Analysis of Turbid Media in the Reflectance Probe Regime". *Quantum Electron*. 2014. 44(7): 641–645. DOI:10.1070/QE2014v044n07ABEH015481.

55. C. Yeong and S. Torquato. "Reconstructing Random Media". *Phys Rev E*. 1998. 57(1): 495–506. DOI:10.1103/PhysRevE.57.495.
56. M. Viana, P. Jouannin, C. Pontier et al. "About Pycnometric Density Measurements.". *Talanta*. 2002. 57(3): 583–93. DOI:10.1016/s0039-9140(02)00058-9.
57. F. Chauchard and S. Roussel. "Sonde spectroscopique et procédé pour détecter une inhomogénéité". WO2010136732 A1, Filed: 2010, Issued: 2010.
58. H. Dehghani, F. Leblond, B. W. Pogue et al. "Application of Spectral Derivative Data in Visible and Near-infrared Spectroscopy.". *Phys Med Biol*. 2010. 55(12): 3381–3399. DOI:10.1088/0031-9155/55/12/008.
59. F. Chauchard, J. Revel and M. Zeaiter. "SAM-Spec® Spectroscopy used to Measure Simultaneously API Concentration and Particle Size Distribution Example of Application on Drug Layering Process.". In *Chimiométrie 2009 Conference*. Paris, 2009. .
60. M. Boiret and F. Chauchard. "Use of Near-Infrared Spectroscopy and Multipoint Measurements for Quality Control of Pharmaceutical Drug Products". *Anal BioanalChem*. 2017. 409(3): 683–691. DOI:10.1007/s00216-016-9756-9.
61. Z. Shi and C. A. Anderson. "Scattering Orthogonalization of Near-infrared Spectra for Analysis of Pharmaceutical Tablets". *Anal Chem*. 2009. 81(4): 1389–1396. DOI:10.1021/ac802105v.
62. G. Kattawar. "A Three-parameter Analytic Phase Function for Multiple Scattering Calculations". *J Quant Spectrosc Radiat Transfer*. 1975. 15: 839–849. DOI:10.1016/0022-4073(75)90095-3.
63. G. Fournier and J. L. Forand. "Analytic Phase Function for Ocean Water". In *Proceedings of SPIE , Ocean Optics XII*, volume 2258. Bergen, 1994. pp. 194–201. DOI:10.1117/12.190063.
64. A. A. Kokhanovsky (editor.) *Light Scattering Reviews 3 - Light Scattering and Reflection*. Chichester, United Kingdom: Praxis Publishing, 2008. ISBN 978-3-540-25315-0. DOI:10.1007/3-540-37672-0.
65. I. Sassi and M. Khemiri. "Scattering of Light from Rough Surfaces, the Limits of Validity of Geometric Optics Approximation Method". In *Progress in Eletromagnetics Research Symposium Proceedings*. Taipei, 2013. pp. 623–625.
66. A. Watanabe. "A Trial Production of a Table of the Optical Crystallographic Characteristics of Crystalline Drugs Including Crystal Habits (Study of Crystalline Drugs by Means of a Polarizing Microscope. XIX)". *Yakugaku Zasshi*. 2002. 122(8): 595–606. DOI:10.1248/yakushi.122.595.
67. N. Sultanova, S. Kasarova and I. Nikolov. "Dispersion Properties of Optical Polymers". *Acta Phys Pol A*. 2009. 116(4): 585–587. DOI:10.12693/APhysPolA.116.585.
68. M. N. Pons, K. Milferstedt and E. Morgenroth. "Modeling of Chord Length Distributions". *Chem Eng Sci*. 2006. 61(12): 3962–3973. DOI:10.1016/j.ces.2006.01.036.
69. A. Mazzolo and B. Roesslinger. "Monte-Carlo Simulation of the Chord Length Distribution Function across Convex Bodies, Non-Convex Bodies and Random Media". *Monte Carlo Methods and Applications*. 2004. 10(3-4): 443–454. DOI:10.1515/mcma.2004.10.3-4.443.
70. A. Ruf, J. Worlitschek and M. Mazzotti. "Modeling and Experimental Analysis of PSD Measurements through FBRM". *Part Part Syst Char*. 2000. 17(4): 167–179. DOI:10.1002/1521-4117(200012)17:4.
71. S. Hansen. "Estimation of Chord Length Distributions from Small-angle Scattering Using Indirect Fourier Transformation". *J Appl Crystallogr*. 2003. 36(5): 1190–1196. DOI:10.1107/S0021889803014262.
72. P. D. Bradley. *The Development of a Novel Silicon Microdosimeter for high LET radiation therapy*. Phd thesis, University of Wollongong, 2000.
73. D. Stoyan, A. Wagner, H. Hermann et al. "Statistical Characterization of the Pore Space of Random Systems of Hard Spheres". *J Non-Cryst Solids*. 2011. 357(6): 1508–1515. DOI:10.1016/j.jnoncrysol.2010.12.033.
74. N. Reboul, E. Vincens and B. Cambou. "A Statistical Analysis of Void Size Distribution in a Simulated Narrowly Graded Packing of Spheres". *Granular Matter*. 2008. 10(6): 457–468. DOI:10.1007/s10035-008-0111-5.
75. A. Mazzolo. "Chord Length Distributions in Dense Stochastic Media". In *First International Conference on Physics and Technology of Reactors and Applications*. Marrakech, 2007. .
76. W. Ji and W. R. Martin. "Determination of Chord Length Distributions in Stochastic Media Composed of Dispersed Microspheres". In *Trans. Am. Nucl. Soc.* 2007. pp. 467–469.
77. Y. G. Shkuratov and Y. S. Grynko. "Light Scattering by Media Composed of Semitransparent Particles of Different Shapes in Ray Optics Approximation: Consequences for Spectroscopy, Photometry, and Polarimetry of Planetary Regoliths". *Icarus*. 2005. 173(1): 16–28. DOI:10.1016/j.icarus.2003.12.022.
78. O. Scheibelhofer, S. Sacher and J. G. Khinast. "Near Infrared Light Spectroscopy: Applicability and new Developments". *European Pharmaceutical Review*. 2014. (1): 51–55.
79. P. D. Foote. "Illumination From a Radiating Disk". *Bulletin of the Bureau of Standards*. 1915. 12: 583–586.
80. M. J. Weber (editor.) *Handbook of Laser Science and Technology - Supplement 2: Optical Materials*. Boca Raton: CRC Press, 1995.
81. H. Kogelnik. "On the Propagation of Gaussian Beams of Light Through Lenslike Media Including those with a Loss

or Gain Variation". *Appl Opt*. 1965. 4(12): 1562. DOI: 10.1364/AO.4.001562.

82. L. Lin and M. Zhang. "Evaluation of Scaling Monte Carlo Methods for Backscattering Properties of Turbid Media with Gaussian Incidence". *Opt Commun*. 2015. 336: 134–139. DOI: 10.1016/j.optcom.2014.09.058.

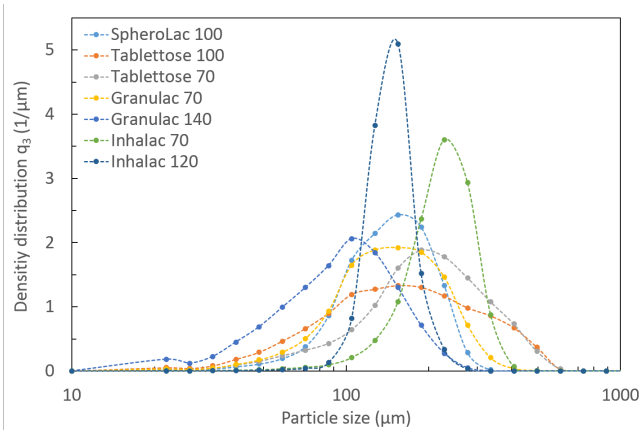


Figure 7. Particle size distributions of the investigated substances.

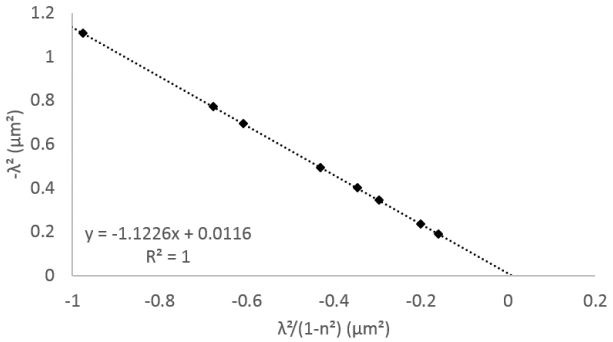


Figure 8. Determination of Sellmeier coefficients from experimental data for refractive index and wavelength.

Supplemental Material

This section describes some of the data and procedures, which are necessary, but would be too extensive to be covered in the main document.

Particle size distribution of investigated materials

Particle size distributions were determined via a QicPic system (Sympatec GmbH, Clausthal-Zellerfeld, Germany). A visualization of the distributions can be found in Figure 7, and the summary parameters are listed in Table 1.

Fitting the Sellmeier dispersion equation

The Sellmeier dispersion equation describes the change of refractive index with changing wavelength. To obtain the necessary coefficients a fit on experimental data, providing refractive index over wavelength for discrete points,[67] was performed. This was done by linearizing a simple form of the dispersion equation

$$n^2 = 1 + \frac{B_1 \lambda^2}{\lambda^2 - C_1} \tag{14}$$

to

$$-\lambda^2 = B_1 \left(\frac{\lambda^2}{1 - n^2} \right) + C_1 \tag{15}$$

, which allows to obtain the coefficients B_1 and C_1. The graphical representation of this procedure is given in Figure 8.

A priori chord placement probability

To determine chord length distributions of individual particles, MC-simulations were used. Determining the length of specific chords inside a sphere is easy, however the distribution is determined based on the a-priory positioning of the chords. This is well known as Bertrand paradoxon in the 2-dimensional case and holds also for the 3-dimensional case. Thus the initial positioning probabilities have to be chosen carefully.

The maximum ignorance principle introduced by Jaynes provides a scale- and translation-invariant solution.[83] This method also corresponds to physical systems, exhibiting these invariants. Moreover, light intensity should follow Lamberts cosine law, i.e., the light intensity on a surface is proportional to the cosine between incoming radiation and surface normal. As such, a uniform position of photons can be chosen on the projected area of an object to represent the object being hit by a beam of parallel rays (the so called μ-random distribution).[84,85] For photon scattering on a sphere with diameter D, the initial photon positions $\mathbf{r}$ are given by

$$\mathbf{r} = \begin{pmatrix} -\frac{D}{2} < x < \frac{D}{2} \\ -\frac{D}{2} < y < \frac{D}{2} \\ Z = -\frac{D}{2} + \varepsilon \end{pmatrix} ; \; x^2 + y^2 < \left(\frac{D}{2} \right)^2 \tag{16}$$

with ε being small. The same mechanism was applied to cubes. However, due to the complexity, no analytical formula was derived. Instead, first, the cube was randomly rotated and, second, a photon position was determined in the projected area of the cube in the $z = -2$ plane.

Chord length distribution in a refracting sphere

For the length distribution of rays passing through a sphere with refractive index n an analytical expression can be derived. Assuming that a ray is parallel to an arbitrary axis through the sphere center and at distance r to the said axis, it will encounter the sphere surface at an angle $\sin(\alpha) = 2r/D$. According to the laws of geometric optics, the ray is refracted to an angle $\sin(\beta) = 1/n \sin(\alpha)$. The path length

inside the sphere is given by

$$s = D\cos(\beta) = D\sqrt{1 - \frac{4r^2}{D^2 n^2}}; \quad \frac{2r}{D} \leq 1 \tag{17}$$

The probability of a ray at distance r (for μ-uniform distributed rays) is given by $p(r) = (2\pi r)/(2\pi D)$. Hence, denoting the cumulative probability as P gives

$$P(s < s') = P\left(r > \frac{n}{2}\sqrt{D^2 - s'^2}\right) = \frac{\int_{\frac{n}{2}\sqrt{D^2-s'^2}}^{\frac{D}{2}} \frac{r}{D}\ dr}{\int_0^{\frac{D}{2}} \frac{r}{D}\ dr} \tag{18}$$

$$= 1 - \frac{n^2}{D^2}\left(D^2 - s'^2\right) \tag{19}$$

which leads to the probability density for the chord lengths

$$q_D(s) = s\frac{n^2}{D^2}; \quad s > \sqrt{\frac{n^2 - 1}{n^2}} \tag{20}$$

Although no analytical solution was derived for cubes, the equations were modified from the existing analytical solution for $n = 1$ to reflect the Monte-Carlo calculations, and are listed as Equation 3.

Chord length in spherical and cubic particles and in voids

The calculations of path lengths is extensively described in the main document. For completeness the distribution of path lengths compared between spherical and cubic particles of same size is shown in graphically. Additionally, the shift in chord length distribution in voids, when porosity is varying is also shown in Figure 9. It is evident, that such changes in the chord length probability distribution lead to different results in the Monte-Carlo simulations.

Geometric and accurate scattering on single particles

In this work, we rely only on geometrical optics. With that regard, it is useful to compare the resulting scattering patterns with the patterns obtained by descriptions including photon phase and interference. In Figure 10, the scattering pattern for a sphere with $x = 200$ (which is the range investigated in this work) obtained via geometric optics is shown. For comparison purposes, intensities obtained via Mie-theory are shown. To calculate the Mie scattering pattern, the script of Schäfer et al. was used.[86,87]

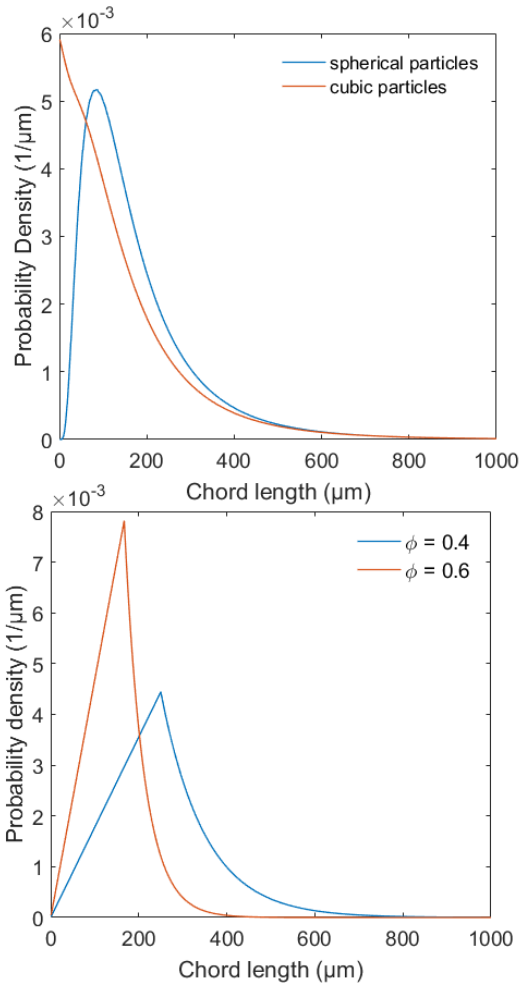


Figure 9. Top: Chord length distribution for spherical particles with diameter D and cubic particles with side length D, assuming a log-normal particle size distribution with $D_{50} = 100\mu$m and $D_{90} = 200\mu$m and a refractive index of 1.46 in both cases. Bottom: Void chord length probability distribution for particles with mean diameter $D = 100\mu$m and two different porosity values.

Chord length and object based scattering

In object-based scattering, the object positions are fixed and rays follow the paths predetermined by geometrical optics. These paths are not completely distinct, since rays still have certain path elements that are determined by chance (e.g., absorption and transmission). In chord length-based scattering, rays take steps whose lengths are determined by a given probability distribution and then encounter stochastically a scattering or absorption event.

Since with samples that are large enough peculiarities in the objects position should even out, both methods should lead to the same result. A brief comparison is shown in Figure 11. Spheres with $n = 1.5$ are distributed uniformly on a 2-dimensional plane. Light enters at the edge and follows the geometric path. Already the few scattering objects lead to a strong diffusion of the incoming ray. This is shown more clearly in the center of Figure 11, where the outgoing ray

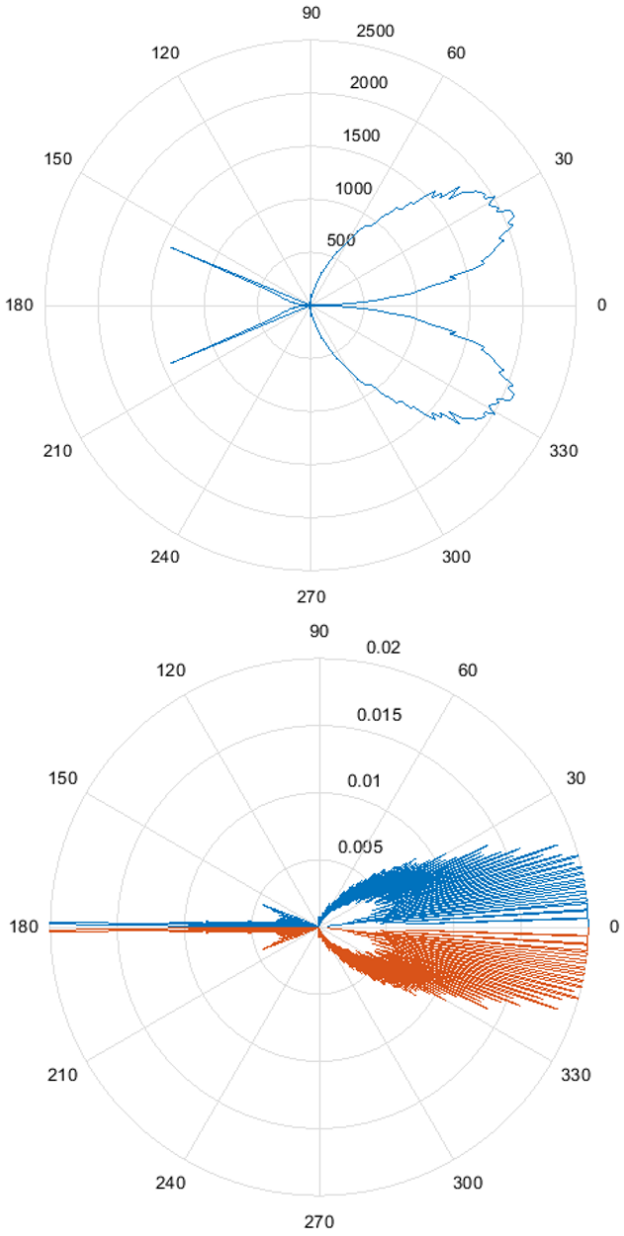


Figure 10. Change in direction of photons scattered over a single sphere with $n = 1.5$ and a dimensionless size factor of $x = 200$. Top: 100,000 photons were used for the geometrical optics description. Bottom: Scattering coefficient of unpolarized light, according to Mie theory. Note that the front lobes are oscillating strongly and extend beyond the shown plot region. However, the overall pattern, exhibiting large forward facing lobes at about 25° is in agreement between both methods.

directions are presented in a polar diagram. The distribution of the recorded individual step lengths of the paths is also provided in Figure 11. The lengths inside the spheres follow the expected chord distribution, and the lengths in between the spheres have Poisson behavior. The obtained chord distributions were used as a basis for a chord-length based simulation within the same sample volume, resulting in the ray paths shown in Figure 12. The two results agree well. Furthermore, it is realized that already in such a diluted sample the reflected radiation is more intense than the transmitted radiation.

Absorption characteristics of α-lactose monohydrate

Absorption was modelled by a sum of Gaussian peaks. Those are described by a certain height $A_i/(\sqrt{2\pi}\sigma_i)$, a full width at half maximum $2\sqrt{2\ln(2)}\sigma_i$ and the peaks center at the known wavelengths $\hat{\lambda}_i$:

$$\epsilon(\lambda) = \sum_i \frac{A_i}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{\left(\lambda - \hat{\lambda}_i\right)^2}{2\sigma_i^2}\right) \qquad (21)$$

Near-infrared peak assignments of lactose can be found in[88–91].

The width and intensity of peaks were adjusted to fit a previously-recorded spectrum, as shown in Figure 13. Note that this spectrum, unlike all others in this work, was acquired using an FT-based spectrometer (Perkin Elmer, Spectrum 400, Waltham, USA) to achieve a higher spectral resolution. Since the experimental spectrum is the sum of absorbed and scattered light, additionally to the absorption part, a polynomial of 2nd order was added to mimic the scattering behavior. This polynomial is not part of the MC calculation. The necessary parameters of the absorption peaks were determined once and kept constant. The individual absorption peaks and their sum in comparison with the recorded spectrum are shown in Figure 13. The individual band parameters are provided in Table 3.

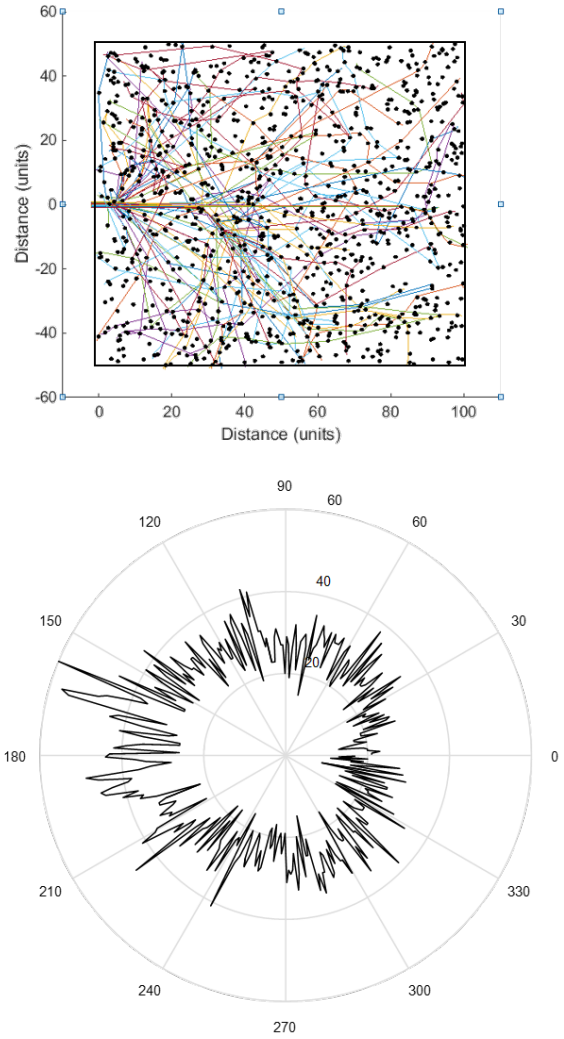
Photon energy distribution

As the white reference is assumed, the number of photons detected can be used straightforward to calculate absorbance, as already described in the main document, at Equation 13. The number of photons emitted, compared to the number of photons detected is shown in Figure 14.

This can now be compared to the photons experimentally determined, when measuring a white reference, and a typical powder bed (in this case SpheroLac 100) as shown in Figure 15.

The resulting absorbance, based on the simulated data, and compared to the experimentally determined absorbance is shown in the main manuscript in Figure 5.

Furthermore, the cause of the increased absorption, i.e. the increasing path length is shown in Figure 16. Evidently, not only is the path length much more than the actual distance, but also does not scale linearly with distance. This mirrors the increasing penetration depth with larger sampling



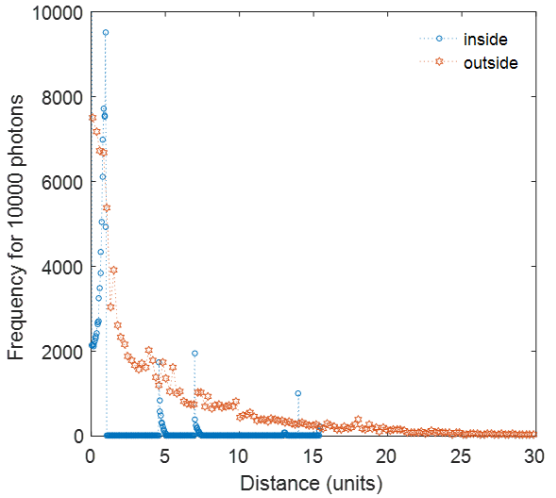


Figure 11. 2-dimensional representation of light scattering. Rays arrive in parallel from a region on the left (with $-1 < y < 1$). They scatter over 1,000 circles with $n = 1.5$ and $D = 1$ that are randomly distributed in a region the size of $100 \cdot 100$. Rays leaving the sample are only drawn for a length of 1. 10,000 rays were used. Top: Paths of light (only 100 of 10000 paths). Center: Direction of the rays leaving the sample. Bottom: Ray path length between the spheres (outside) and inside the spheres (inside).

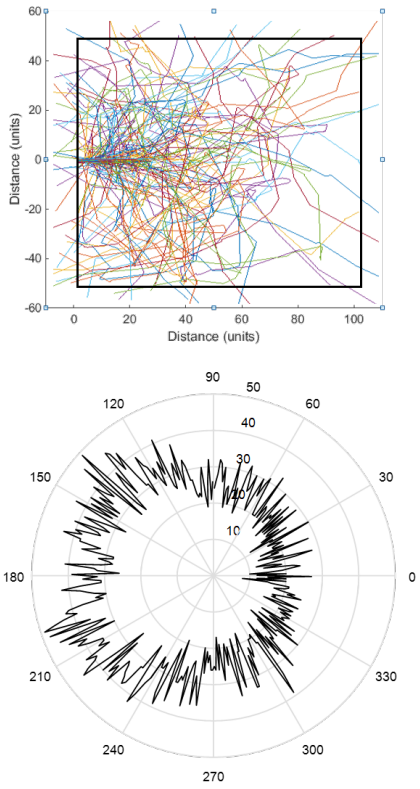


Figure 12. 2-dimensional light scattering. Rays arrive in parallel from a region on the left (with $-1 < y < 1$). The photons follow probabilistic steps and encounter randomly oriented interfaces. Top: The photon paths for 100 of the 10,000 photons. Bottom: The final photon direction. Compare the emerging pattern to that in Figure 11.

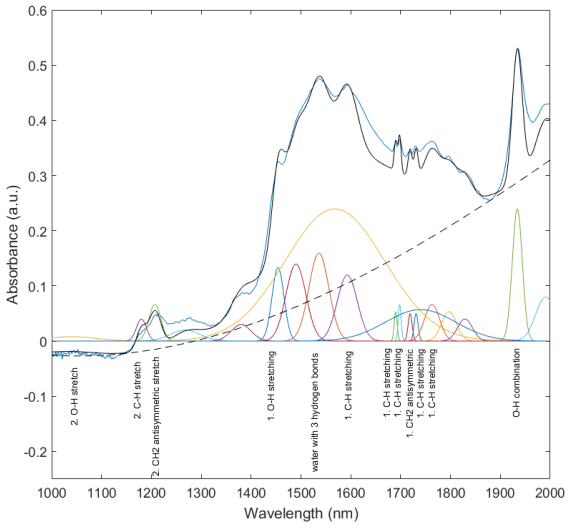


Figure 13. Individual absorption bands of α-lactose monohydrate, as stated in Table 3. The sum of individual peaks with an additional 2nd order polynomial background (black dashed line) is shown as a black solid line. The experimentally acquired spectrum is shown in blue.

distance. The travelled distance also varies amply from photon to photon, and the standard deviations are marked in

Table 3. Parameter settings of individual bands for modelling absorbance: the corresponding wavelength λ, the relative absorption strength A and the broadness of peak σ. Assignment to a specific vibration mode is given, if feasible.

λ (nm)	A	σ (nm)	Assignment
926	0.8	1	C-H stretch
1040	1	50	2 O-H stretch
1180	1.1	10	2 C-H stretch
1207	1.5	12	2 CH_2 antisymmetric stretch
1268	1	30	
1380	2	20	
1454	4	12	1 O-H stretching
1490	7	20	
1536	8	20	water with 3 hydrogen bonds
1568	60	100	overlapping of a large number of smaller bonds
1593	6	20	1 C-H stretch
1690	0.4	3	1 C-H stretch
1698	0.5	3	1 C-H stretch
1719	0.5	4	1 CH_2 antisymmetric stretch
1731	0.5	4	1 C-H stretch
1740	10	70	
1763	2	15	1 C-H stretch
1797	1	15	
1829	0.5	15	
1934	5	10	O-H combination
1991	2	15	

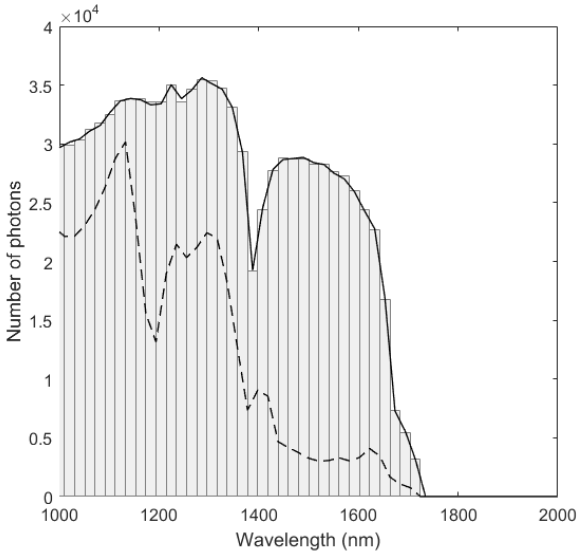


Figure 14. The solid line gives the initial photon probability per wavelength (for 1 million photons), as determined from experimental measurements of white standard. The bars represent the actual number of binned photons emitted in simulation. Bin size is 20 nm. The dashed line represents the finally collected photon intensity for the different wavelengths.

Figure 16 with error bars. Of course, negative travel distances are not possible, and the mean travelled path length does not follow a normal distribution.

Additional references

83. E. T. Jaynes. "The well-posed problem". *Found Phys*. 1973. 3(4): 477–492. DOI:10.1007/bf00709116.

84. K. Eckerman, J. Ryman, A. Taner et al. "Traversal of Cells by Radiation and Absorbed Fraction Estimates for Electrons and Alpha Particles". In A. T. Schlafke-Stelson and E. E. Watson (editor.) *Proceedings of the fourth international radiopharmaceutical dosimetry symposium*. Oak Ridge: Oak Ridge Associated Universities, 1986. pp. 26–36.

85. N. Sjöstrand. "What is the Average Chord Length?". *Ann Nucl Energy*. 2002. 29(13): 1607–1608. DOI:10.1016/S0306-4549(02)00003-8.

86. J.-P. Schäfer. *Implementierung und Anwendung analytischer und numerischer Verfahren zur Lösung der Maxwellgleichungen für die Untersuchung der Lichtausbreitung in biologischem Gewebe*. PhD Thesis, University Ulm, 2011.

87. J. Schäfer, S. C. Lee and A. Kienle. "Calculation of the Near Fields for the Scattering of Electromagnetic Waves by Multiple Infinite Cylinders at Perpendicular Incidence". *J Quant Spectrosc Radiat Transfer*. 2012. 113(16): 2113–2123. DOI:10.1016/j.jqsrt.2012.05.019.

88. Á. Gombás, I. Antal, P. Szabó-Révész et al. "Quantitative Determination of Crystallinity of alpha-Lactose Monohydrate by Near Infrared Spectroscopy (NIRS)". *Int J Pharm*. 2003. 256(1-2): 25–32. DOI:10.1016/S0378-5173(03)00059-0.

89. H. P. Aenugu, D. S. Kumar, Srisudharson et al. "Near Infra Red Spectroscopy - An Overview". *Ing J ChemTech Res*. 2011. 3(2): 825–836.

90. M. Sekkal and P. Legrand. "A Vibrational Molecular Force Field of Model Compounds with Biological Interest - III. Harmonic Dynamics of α- and β-D-galactose in the Crystalline State". *Spectrochim Acta*. 1992. 48(7): 959–973. DOI:10.1016/0584-8539(92)80172-S.

91. H. M. Gaffour, M. Bouterfas, N. Bekhti et al. "Harmonic Dynamics of α-D-Lactose in the Crystalline State". *J Mol Imaging Dyn*. 2011. 01(01): 1000102. DOI:10.4172/2155-9937.1000102.

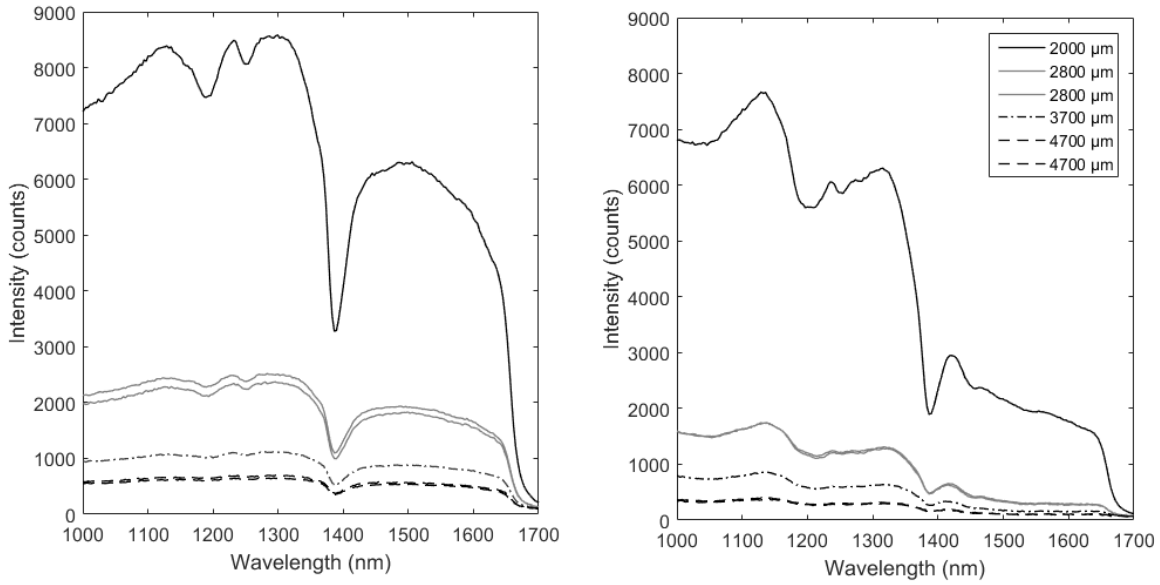


Figure 15. Measured light intensity for the considered wavelength range, for white reference material (left) and SpheroLac 100 (right).

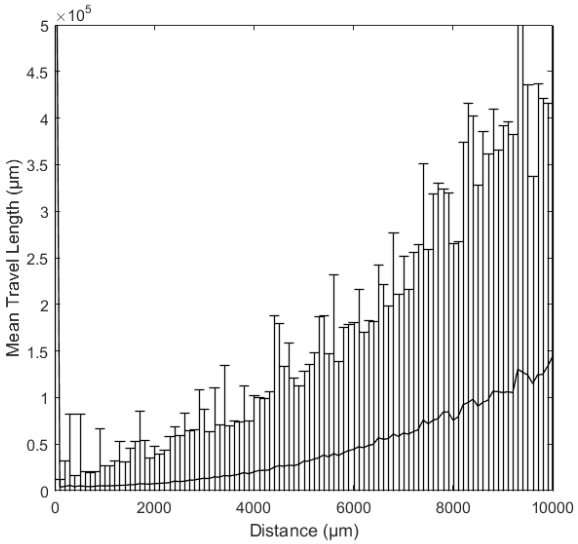


Figure 16. Mean path length in dependence of photon exit point, for parameters for SpheroLac 100. Roughly, the effective photon path length inside material particles is 100 times the distance from the illumination spot center.